Ex: 9
11/3/26

EXPERIMENTAL DETERMINATION OF Z PARAMETERS.

AIM:

To Calculate and Verify Z Parameters of a two Port Network using Multisim Simulator.

APPARATUS REQUIRED:

S.No	COMPONENTS	SPECIFICATION	QUANTITY
1.	Power Supply	–	1
2.	Multimeter	10KΩ, 10KΩ, 4.7KΩ	–
3.	Resistors	–	1
4.	Bread Board	–	1
5.	Connecting wires	–	–
6.	PC with Multisim Simulator	–	–

OBSERVATION:

S.No	OUTPUT OPEN CIRCUITED ($I_2=0$)			$Z_{21} = V_2/I_1$ (Ω)	$Z_{11} = V_1/I_1$ (Ω)
	V_1	V_2	I_1		
1.	5.0	2.60	2.10	1.23kΩ	2.38kΩ
2.	5.0	2.55	2.08	1.22kΩ	2.40kΩ
3.	5.0	2.63	2.12	1.24kΩ	2.35kΩ

S.No	INPUT OPEN CIRCUITED ($I_1=0$)			$Z_{12} = V_1/I_2$ (Ω)	$Z_{22} = V_2/I_2$ (Ω)
	V_2	V_1	I_2		
1.	5.0	2.60	2.10	1.23	4.00
2.	5.0	2.57	2.09	1.22	2.39
3.	5.0	2.62	2.12	1.24	2.38.

OK verified

PROCEDURE:

a) Connect the circuit as shown in Fig. & Switch 'ON' the experimental board.

b) First open the o/p terminal & supply 5V to I/p terminal. Measure o/p Voltage and I/p Current.

c) Secondly, open I/p terminal & supply 5V to o/p terminal. Measure I/p Voltage & o/p Current.

d) Switch 'OFF' the Supply after taking the readings.

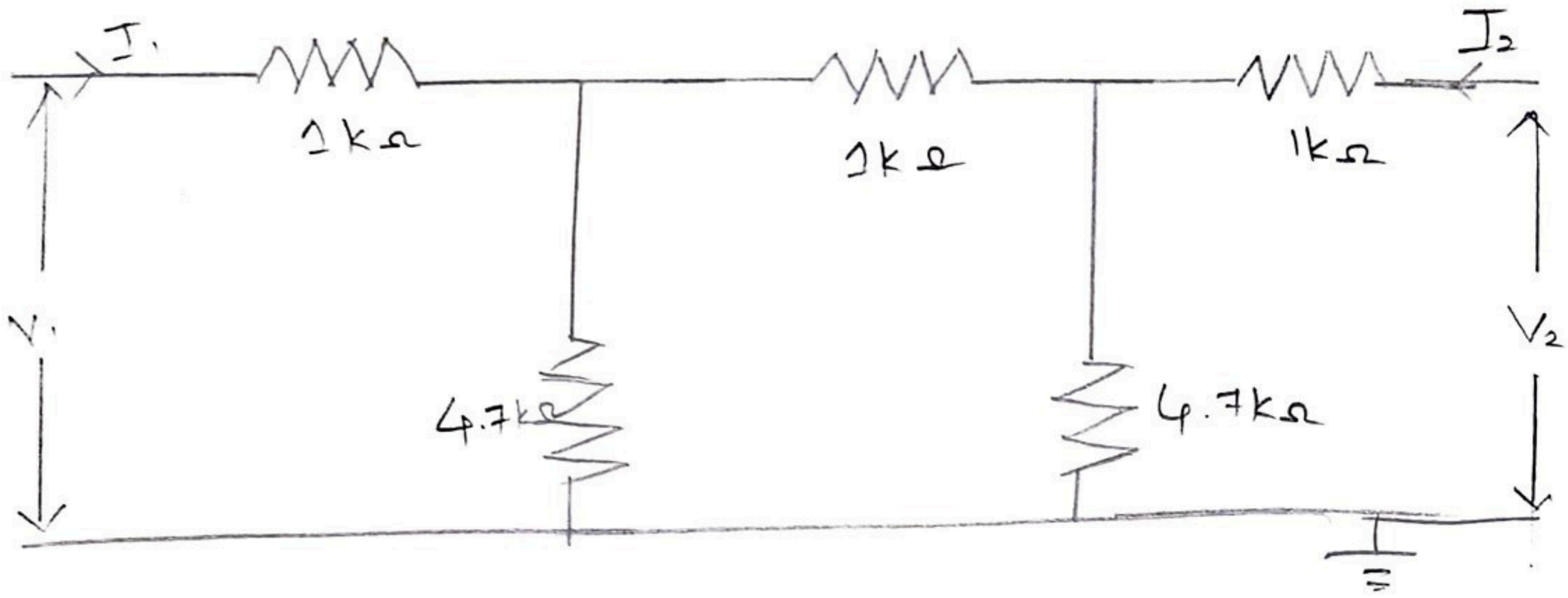
MARKS SPLIT-UP:

S.No	Description	Marks allotted	Marks obtained.
1.	Experimental setup	20	20
2.	observation/Readings	20	20
3.	Calculations/Results	20	20
4.	Viva	20	20
5.	Record	20	20
		Total	100

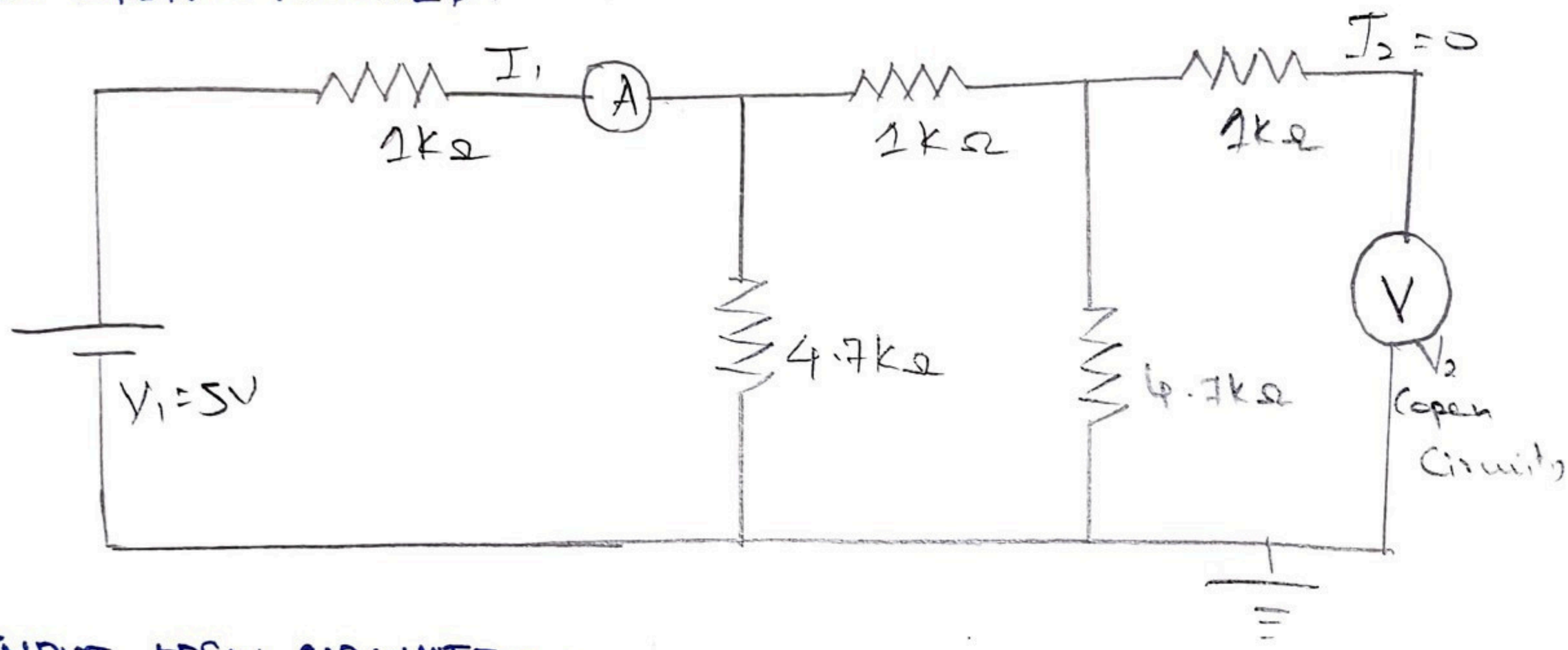
RESULT:

The Z-Parameters of the two Port network has been Calculated & Verified by using Multisim simulator.

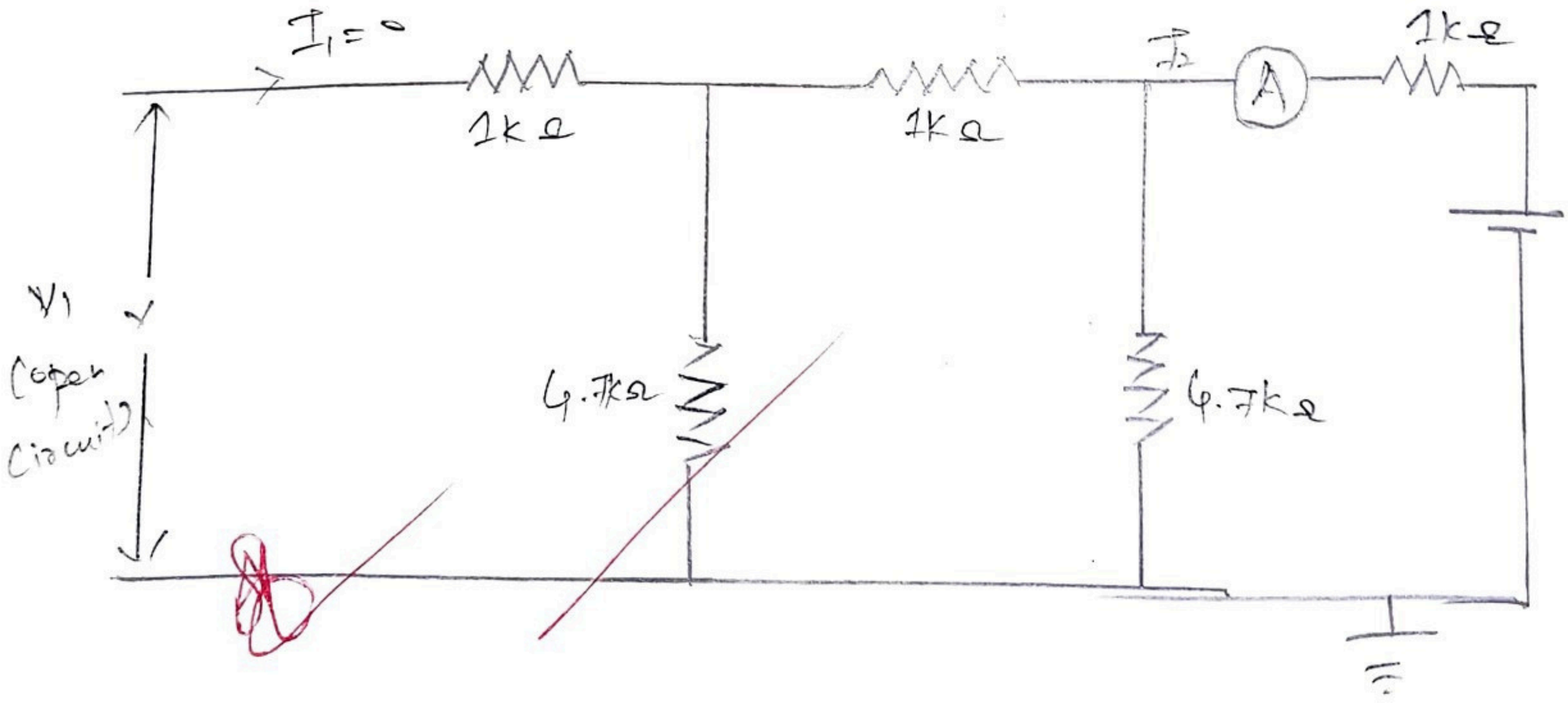
CIRCUIT DIAGRAM:

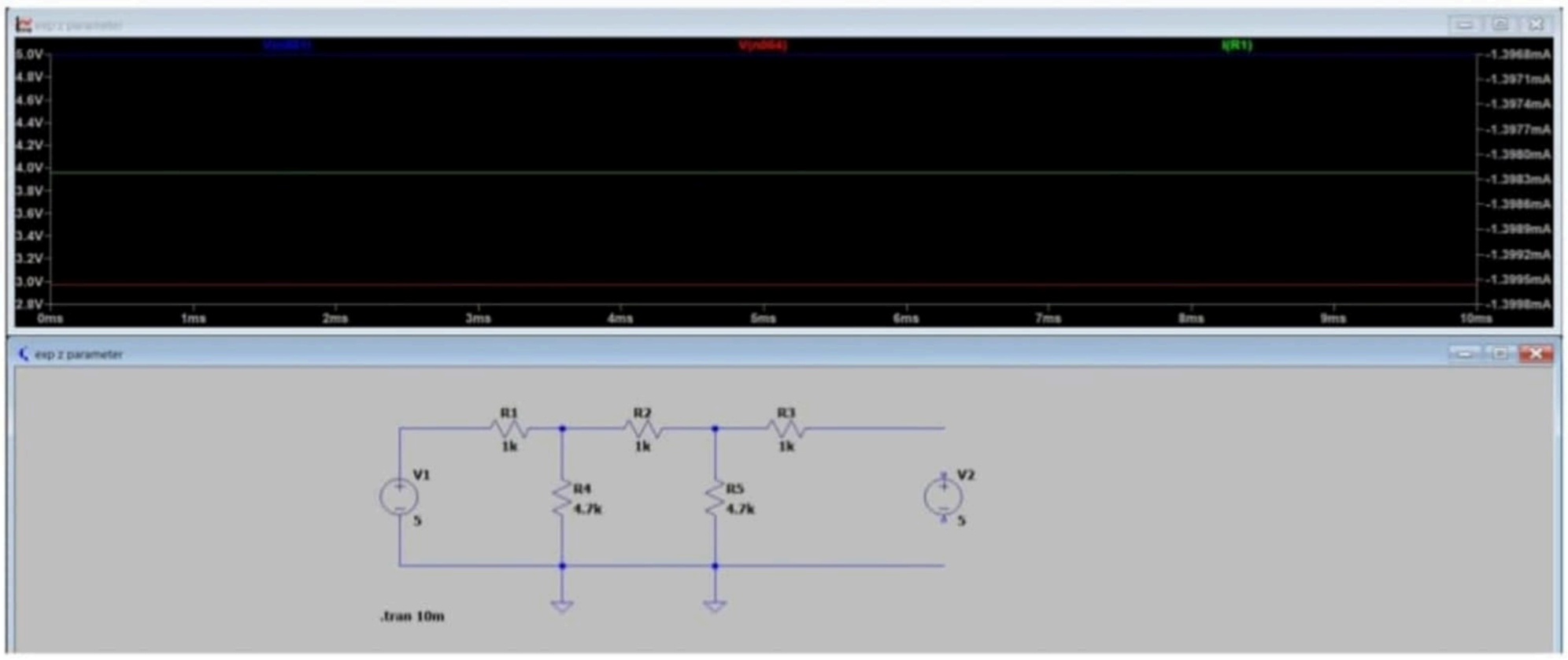


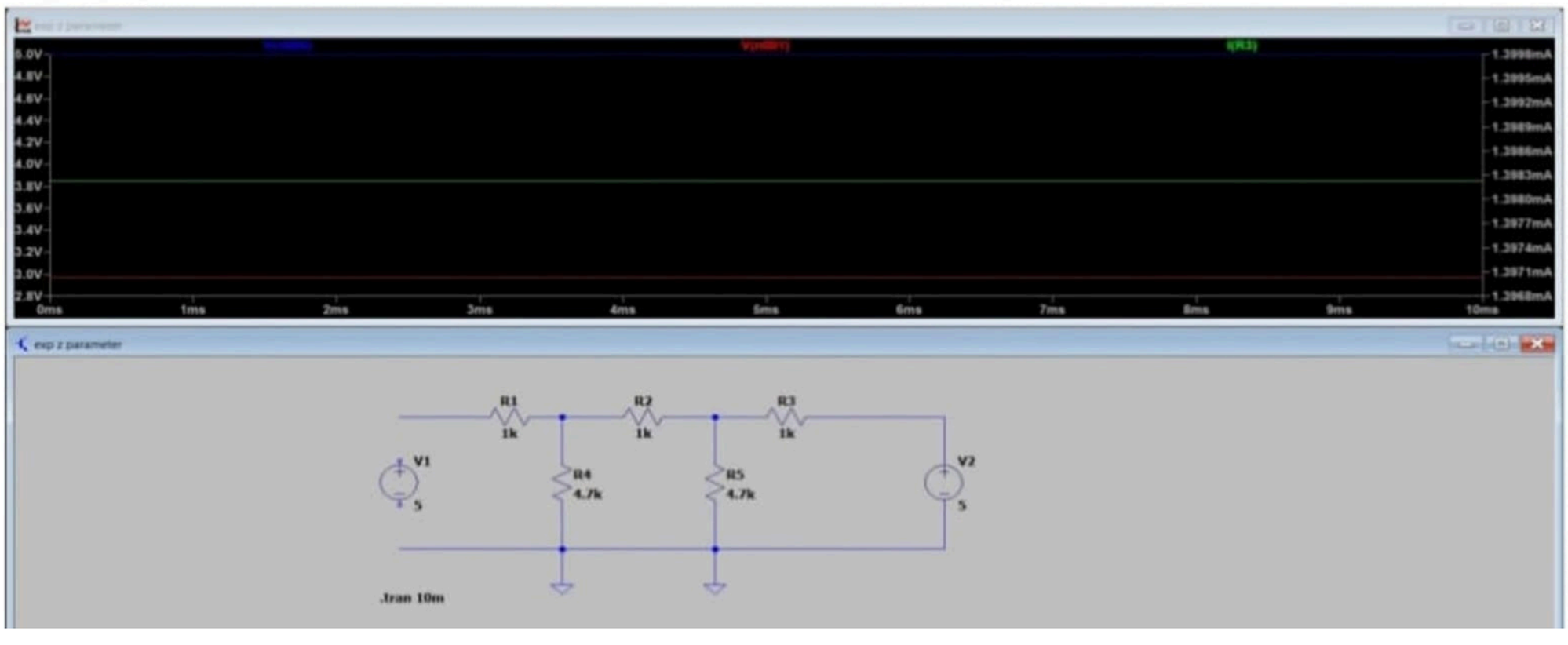
OUTPUT OPEN CIRCUITED:



INPUT OPEN CIRCUITED:




exp z parameter
I(R1)
R1
1k
R2
1k
R3
1k
V1
5
R4
4.7k
R5
4.7k
V2
5
.tran 10m


exp z parameter
I(R3)
R1
1k
R2
1k
R3
1k
V1
5
R4
4.7k
R5
4.7k
V2
5
.tran 10m